

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 48

Enforce IMDSv2 with an AWS Organizations EC2 Declarative Policy

AWS Organizations | EC2 Policies | IMDSv2 | S3 Status Reports

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
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STATUS	Complete

Executive Summary

Created an organization-level configuration baseline using **AWS Organizations declarative policies for EC2**, which the current console labels as **EC2 policies**. Working from the management account, an S3 report bucket was created in **us-east-1** and configured with a resource policy allowing the declarative-policy reporting service to write status results. The EC2 policy type was enabled, an organization status report was generated, and the report showed uniform settings with **zero inconsistent accounts**. A customer-managed policy named **Instance-Meta-Data-Service-v2** was then created to require IMDSv2 defaults and attached to the **Prod OU**.

SCOPE CLARIFICATION

This policy enforces EC2 instance metadata defaults for new launches in the attached scope. It is a desired-state configuration control, not an IAM permission policy and not a retroactive reconfiguration of every existing instance.

Business Problem

Permission policies control who can perform actions, but they do not guarantee that every account maintains the same security configuration. Before declarative policies, organizations commonly deployed defaults through Infrastructure as Code and used SCPs to restrict changes. Declarative policies provide a centralized way to define and continuously maintain supported EC2 security settings across accounts and organizational units.

Objectives

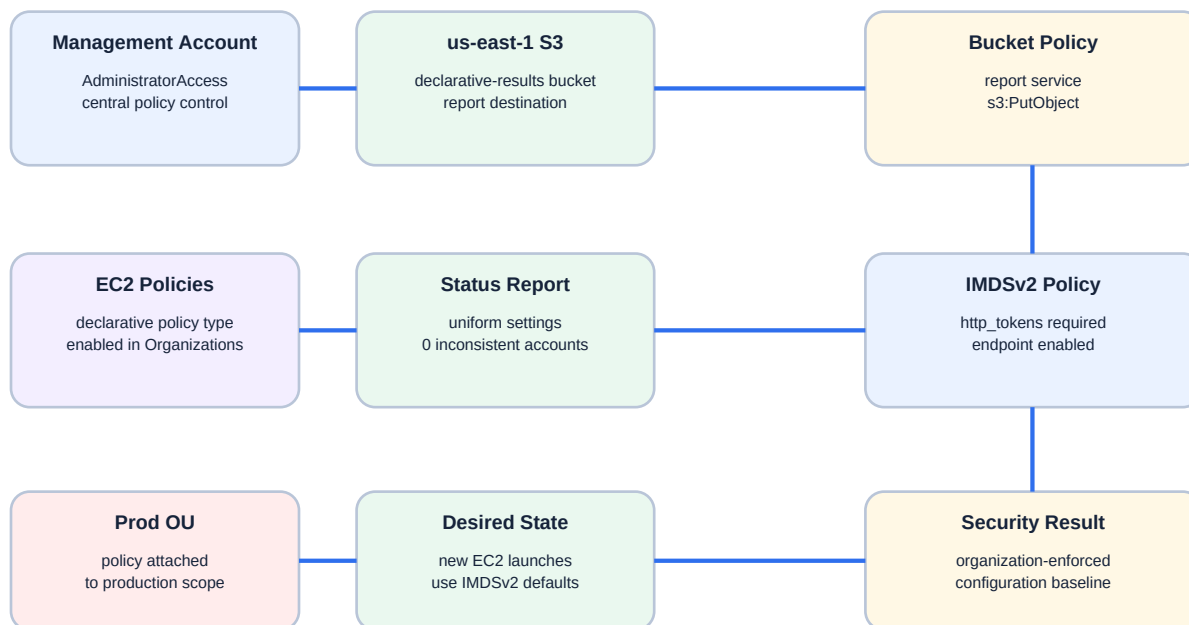
- Sign into the AWS Organizations management account.
- Create an S3 bucket in us-east-1 for declarative-policy status reports.
- Add a JSON bucket policy allowing the EC2 declarative-policy report service to write objects.
- Open AWS Organizations and enable the declarative policy type now presented as EC2 policies.
- Generate an account status report scoped to the organization.
- Review the report for conflicts and inconsistent account settings.
- Create a customer-managed EC2 policy for Instance Metadata Defaults.
- Enable the instance metadata service and require IMDSv2 token usage.
- Attach the policy to the Prod organizational unit.

Environment

Cloud Provider	Amazon Web Services
Administrative Context	AWS Organizations management account / AdministratorAccess
Report Bucket Region	us-east-1 / US East (N. Virginia)

Console Context After Bucket Setup	Returned to us-west-2 before opening the global AWS Organizations console
Policy Type	Declarative policy for EC2 / EC2 policy (customer managed)
Status Report Scope	Organization root
Status Report Result	Uniform across accounts; 0 inconsistent accounts in captured report
Policy Name	Instance-Meta-Data-Service-v2
Description	Only Allow IMDSv2
Service Attribute	Instance Metadata Defaults
Metadata Endpoint	Enabled
Token Requirement	V2 only / token required
Target	Prod organizational unit
Status	Complete

Architecture



Declarative policies enforce configuration state; they do not grant or deny API permissions.

Figure 1 - Status-report workflow and organization-level IMDSv2 configuration enforcement.

Implementation Summary

1. Logged into the AWS Organizations management account.
2. Changed to us-east-1 and opened Amazon S3.
3. Created a new declarative-results report bucket.
4. Edited the bucket policy using JSON so the declarative-policies EC2 reporting service could write report objects.
5. Returned to the normal us-west-2 console context and opened AWS Organizations.
6. Enabled declarative policies for EC2, now displayed in the console as EC2 policies.
7. Opened the account status report workflow and selected the new S3 bucket as the report destination.
8. Scoped the report to the organization root and generated the report.
9. Reviewed the report and found uniform settings with no inconsistent accounts.
10. Created a customer-managed EC2 policy named Instance-Meta-Data-Service-v2.
11. Enabled Instance Metadata Defaults, enabled IMDSv2 enforcement, and selected V2 only (token required).
12. Attached the policy to the Prod OU.

Report Bucket Policy Pattern

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Sid": "DeclarativePoliciesReportBucket",
    "Effect": "Allow",
    "Principal": {
      "Service": [
        "report.declarative-policies-ec2.amazonaws.com"
      ]
    },
    "Action": ["s3:PutObject"],
    "Resource": "arn:aws:s3:::<report-bucket>/*",
    "Condition": {
      "StringEquals": {
        "aws:SourceArn":
          "arn:aws:declarative-policies-ec2:us-east-1:<account-id>:*"
      }
    }
  }]
}
```

EVIDENCE NOTE

The supplied JSON screenshot contains placeholder values. The actual policy must use the created bucket ARN with /* and the management-account ID in the SourceArn condition.

Declarative Policy Content

```

{
  "ec2_attributes": {
    "instance_metadata_defaults": {
      "http_tokens": {
        "@@assign": "required"
      },
      "http_put_response_hop_limit": {
        "@@assign": -1
      },
      "http_endpoint": {
        "@@assign": "enabled"
      },
      "instance_metadata_tags": {
        "@@assign": "no_preference"
      }
    }
  }
}

```

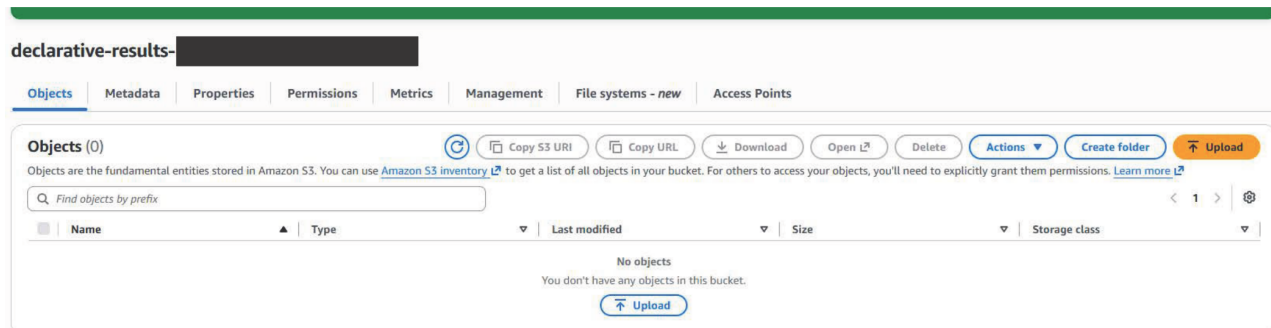
The generated policy content sets **http_tokens** to **required**, keeps the metadata endpoint enabled, leaves metadata tags at no preference, and uses the service-generated hop-limit assignment shown by the policy.

Configuration-Control Comparison

IAM policy	Grants or denies actions for identities	Not used to enforce the IMDS configuration state.
SCP	Sets maximum actions available to accounts and identities	Could restrict changes, but does not define the desired setting itself.
RCP	Sets maximum permissions available to supported resources	Not the mechanism used for EC2 metadata defaults.
Declarative EC2 policy	Defines and enforces supported EC2 configuration state	Requires IMDSv2 defaults for the Prod OU.
Status report	Evaluates current organization configuration consistency	Showed uniform settings and zero inconsistent accounts before deployment.

Evidence

Figure 2 - Declarative Policy Report Bucket Created in us-east-1



The S3 evidence shows the redacted declarative-results bucket with no objects at the time of capture. The bucket was created as the destination for EC2 policy account-status reports.

Figure 3 - Report Bucket Policy JSON Template

```

1  {
2    "Version": "2012-10-17",
3    "Statement": [{
4      "Sid": "DeclarativePoliciesReportBucket",
5      "Effect": "Allow",
6      "Principal": {
7        "Service": ["report.declarative-policies-ec2.amazonaws.com"]
8      },
9      "Action": ["s3:PutObject"],
10     "Resource": "arn:aws:s3:::amzn-s3-demo-bucket/*",
11     "Condition": {
12       "StringEquals": {
13         "aws:SourceArn": "arn:aws:declarative-policies-ec2:us-east-1:your-12-digit-account-id:*"
14       }
15     }
16   }]
17 }
```

The JSON evidence shows the `DeclarativePoliciesReportBucket` statement, the `report.declarative-policies-ec2.amazonaws.com` service principal, `s3:PutObject`, a report-bucket object ARN, and an account-scoped `SourceArn` condition. Placeholder values are visible in this template screenshot.

Figure 4 - Account Status Report Shows Uniform Configuration

Account status report Updated today at 12:44 PM Generate status report View report in S3				
Attribute	Region	Uniform across accounts	Inconsistent accounts	Most frequent value
▶ Allowed Images Settings	All Regions	✔ Yes	0	{"State":"disabled"}
▶ VPC Encryption Control	All Regions	✔ Yes	0	{"State":"default-state",
▶ Instance Metadata Defaults	All Regions	✔ Yes	0	{}
▶ Serial Console Access	All Regions	✔ Yes	0	{"SerialConsoleAccessEn
▶ VPC Block Public Access	All Regions	✔ Yes	0	{"InternetGatewayBlock
▶ Snapshot Block Public Access	All Regions	✔ Yes	0	{"State":"unblocked"}
▶ Image Block Public Access	All Regions	✔ Yes	0	{"ImageBlockPublicAcce

The account status report shows all displayed attributes as uniform across accounts with zero inconsistent accounts. Instance Metadata Defaults displays an empty most-frequent value, establishing that the IMDSv2 default had not yet been configured.

Figure 5 - IMDSv2 Instance Metadata Defaults Configured

Visual Editor

JSON Editor

▼ Instance Metadata Defaults

Remove

Controls IMDS defaults for new EC2 instances.

Instance metadata service

Controls IMDS defaults for new EC2 instances.

Enabled

Enforce IMDSv2

Enabled

Metadata version

V2 only (token required)

⚠ Ensure instances are not making v1 calls

Before setting Metadata version to V2 only (token required), ensure that none of your instances are making V1 calls. The MetadataNoToken CloudWatch metric tracks V1 calls. [View MetadataNoToken](#)

Access to tags in metadata

No preference

Metadata response hop limit

1

Hop limit must be between 1 and 64.

The EC2 policy visual editor shows Instance Metadata Defaults enabled, Enforce IMDSv2 enabled, Metadata version set to V2 only (token required), access to tags in metadata set to No preference, and the service warning to verify that instances are not making v1 calls.

Figure 6 - Customer-Managed EC2 Policy Attached to Prod

Instance-Meta-Data-Service-v2

Delete

Edit policy

Policy details

Name

Instance-Meta-Data-Service-v2

ARN

arn:aws:iam::[redacted]:policy/[redacted]/declarative_policy_ec2-[redacted]

Policy type

EC2 policy (customer managed)

Description

Only Allow IMDSv2

Content

Targets

Tags

Targets

Detach

Attach

Name	ID	Type
☐ Prod	[redacted]	ORGANIZATIONAL_UNIT

The policy details evidence shows Instance-Meta-Data-Service-v2, policy type EC2 policy (customer managed), description Only Allow IMDSv2, and the Prod organizational unit listed as the target. Account and policy identifiers are redacted.

Security Analysis

Declarative policies define and maintain supported service configuration states. They are separate from authorization-policy evaluation.

Requiring IMDSv2 reduces exposure to simple server-side request forgery paths because metadata requests require a session token.

The policy sets account-level metadata defaults for new EC2 instances in the Prod OU. Existing instances should still be reviewed individually.

The account status report is a pre-deployment safety control. It identifies inconsistent values that may signal application compatibility or rollout risk.

The report bucket policy is narrowly scoped to the declarative-policy reporting service, object writes, the report bucket path, and the management-account SourceArn.

Attaching first to Prod is security-forward but operationally sensitive. Production workloads should be checked for IMDSv1 dependencies before enforcing token-required metadata.

Lessons Learned

Declarative policies enforce configuration state rather than API authorization.

The current AWS Organizations console may label declarative policies for EC2 as EC2 policies.

Status reports should be generated before a new organization-wide configuration is attached.

The report bucket must be created in us-east-1 and allow the reporting service to write objects.

A uniform report does not mean the desired control is already enabled; the Instance Metadata Defaults value was still unset.

IMDSv2 enforcement belongs with least-privilege instance roles, application SSRF defenses, and metadata-call monitoring.

Production Improvements

- Review the CloudWatch MetadataNoToken metric before enforcing IMDSv2 broadly.
- Test the policy in a sandbox or NonProd canary account before production rollout.
- Validate existing production instances and launch templates for IMDSv1 dependencies.
- Create a custom error message with a support or remediation reference.
- Automate status-report generation and archive reports with retention and encryption.
- Use CloudTrail and EventBridge to monitor EC2 policy creation, attachment, update, and detachment.
- Expand declarative policies to supported controls such as allowed images, image public access, snapshot public access, serial console, and VPC Block Public Access.
- Re-run the status report after policy deployment and document the effective configuration.

Skills Demonstrated

Declarative / EC2 policies	Account status reports	IMDSv2 defaults
S3 report destination	Service-principal bucket policy	OU targeting
Desired-state enforcement	Pre-deployment validation	Production rollout planning

Resume Bullet

RESUME	Enabled AWS Organizations EC2 declarative policies, created an S3-backed organization status-report workflow, validated uniform account configurations, and attached a customer-managed policy to the Prod OU requiring IMDSv2 token-based metadata defaults for new EC2 launches.
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STAR Interview Story

Situation: The portfolio had demonstrated the risks of IMDSv1 and the limitations of relying on account-by-account configuration.

Task: Create a centrally enforced production baseline for EC2 instance metadata while checking the organization for configuration conflicts first.

Action: I created a us-east-1 S3 report bucket and service-scoped bucket policy, enabled EC2 declarative policies, generated an organization status report, confirmed zero inconsistent accounts, created an IMDSv2 policy, and attached it to the Prod OU.

Result: The production OU gained an organization-enforced IMDSv2 default for new EC2 launches, with report-based pre-deployment validation and evidence of the final OU target.

Version History

v1.0.0	August 6, 2026	Initial documentation. Created the us-east-1 status-report bucket and policy, enabled EC2 declarative policies, generated and reviewed the organization status report, created the IMDSv2 configuration policy, attached it to Prod, and embedded five evidence screenshots.

References

- Cloud Security Lab a Week - Enforce Org-Wide Configurations with Declarative Policies.
- User-provided declarative-results bucket evidence.
- User-provided report bucket policy JSON evidence.
- User-provided account status report evidence.
- User-provided IMDSv2 EC2 policy settings evidence.
- User-provided Prod OU target evidence.
- Project 38 - IMDSv1 and IMDSv2 Metadata Credential Risk.